CODEMOOD · PROJECT SUMMARY

Hinglish Emotion Detection

What the base paper proposed, what we have built, and what we still need to test

Team: CodeMood - Sarcasm-Aware Hinglish Analysis
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We started with a paper that combines text and emoji cues for English and Hinglish sentiment. Its idea is useful, but its own results expose the problem we want to solve: the models work well on English and much less well on Hinglish. Our project keeps that starting point and asks how spelling variation, code mixing, emojis, and sarcasm affect the decision. We are building the system in stages so that each change can be checked on its own.

The starting point

The paper describes BERT and RoBERTa for text, Emoji2Vec for emoji features, and a positive/neutral/negative classifier. It also maps confidence into stronger or weaker emotion wording. The published validation accuracies are 95.7% for both models on English, 59.5% for BERT on Hinglish, and 57.9% for RoBERTa on Hinglish. In the displayed Hinglish confusion matrices, BERT gets 15 negative examples right and RoBERTa gets none. These are results from the base paper, not results from our revised system.

The paper describes text–emoji fusion, although the displayed training code mainly shows standard text-classifier fine-tuning. We therefore treat the described architecture and the demonstrated code as separate things to reproduce and verify.

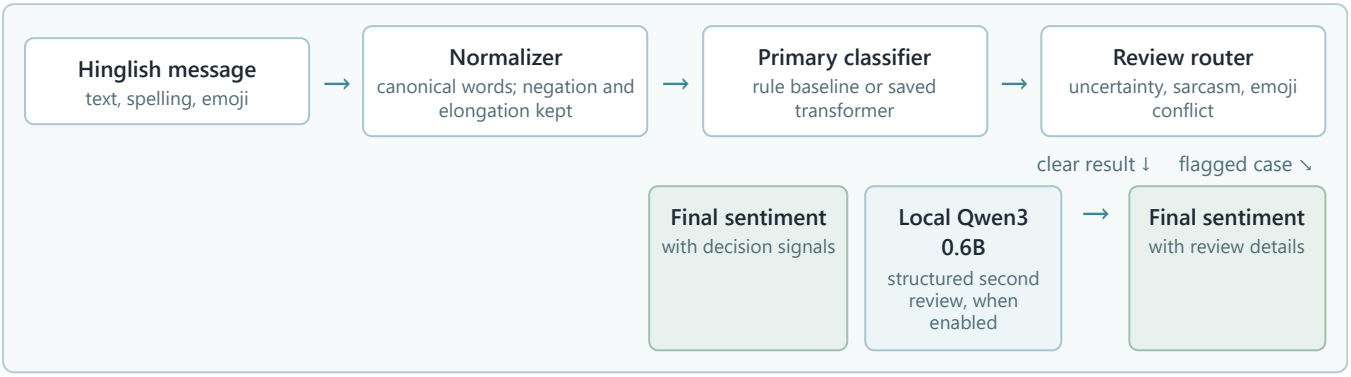
Our five changes

Area	What we are changing	Current position
1. Language model	Compare the English BERT/RoBERTa baselines with MuRIL, mBERT, and XLM-R under the same data split. MuRIL is a sensible first candidate for Romanized Hindi, but the result must decide.	Training path built Comparison pending
2. Hinglish spelling	Map forms such as *acha*, *achha*, and *acchaaa* consistently. Keep a separate marker for elongated letters and preserve negation such as *nahi*.	Normalizer built Robustness study pending
3. Emoji and sarcasm	Read emoji meaning against the words around it. A future supervised model should predict sentiment and sarcasm together; the present prototype only flags likely conflicts and patterns.	Heuristics built Fusion/head pending
4. Local review	Send difficult cases to local Qwen after checking confidence, sarcasm cues, negation, and word–emoji conflict. The installed project default is Qwen3 0.6B, chosen for the available laptop.	Working option Benefit unmeasured
5. Evaluation	Look beyond accuracy: macro-F1, class recall, confusion matrices, confidence calibration, spelling consistency, and tests that should flip sentiment.	Tools built Research dataset needed

What the project does now

The runnable local path and its supporting tools

Architecture of the working prototype



This diagram shows the code path today. MuRIL/XLM-R training and learned emoji fusion belong to the proposed research stage; they are not silently assumed to be running.

Message handling

The normalizer keeps the raw text, a normalized version, token details, detected spelling elongation, and negation. The present rule classifier combines known sentiment words with emoji cues. It is a transparent check of the pipeline, not our final model.

Additional research baselines

A simple intensity module estimates valence and arousal from visible cues. Conversation memory tracks changes for the same speaker across turns. An evidence check masks highlighted words and sees whether the prediction changes. These are separate experiments, not hidden parts of the main classifier.

When review is requested

The router records why it flagged a case: low confidence, possible sarcasm, an emoji conflict, or uncertain negation. If the user enables Ollama, Qwen is asked for a constrained sentiment label, sarcasm flag, emoji relation, and short evidence list.

Dashboard and data

The Streamlit dashboard has overview, text analysis, model effectiveness, and dataset exploration views. It accepts CSV/XLSX, checks labels, plots class balance and live model results, and keeps analysis history in the browser session. The bundled 75-phrase workbook is for demonstrations and smoke tests.

One example of why this matters

Consider “wah kya service hai, 2 ghante late 😂”. A literal reading of *wah* can suggest praise, while the delay points the other way. The current router can flag this clash and ask local Qwen for a second reading. Whether that improves overall accuracy is a question for the held-out evaluation, not something a single example proves.

How we will check the changes

The project is ready for a careful comparison; it is not yet a trained research result

Work still to complete

1. **Reproduce the base paper.** Use a fixed split and seed for BERT and RoBERTa. Report class-level errors, particularly the negative class, before introducing new components.
2. **Train and compare language encoders.** The repository contains a transformer fine-tuning script with a fixed train/validation/test split, optional class weighting, early stopping on validation macro-F1, and saved test metrics. We still need the full labelled dataset and actual runs for MuRIL, mBERT, and XLM-R.
3. **Test spelling changes.** Build a held-out set of equivalent spellings and elongated forms, plus negation examples that must change the label. Compare the raw and normalized inputs rather than assuming normalization helps every case.
4. **Train emoji and sarcasm components.** The earlier proposal called for contextual emoji fusion and a shared sentiment/sarcasm model. Those layers have not been trained. They require reliable sarcasm labels and enough emoji-conflict cases to test them.
5. **Measure local review.** Compare the main classifier with and without routed Qwen review. Count correct changes, wrong changes, review coverage, and response time on the 8 GB CPU-only laptop.

What we will report

Measure	Why it belongs in the report
Accuracy and macro-F1	Overall performance and whether one frequent class hides weak minority-class results.
Per-class precision/recall and confusion matrix	Whether negative, neutral, and positive Hinglish messages are each recognized.
Sarcasm F1 and emoji-conflict cases	Whether the new context handling solves the examples that motivated it.
Variation consistency and sentiment flips	Whether spelling changes keep meaning stable and negation changes it when expected.
Calibration, route rate, and latency	Whether confidence can guide review and what the local reviewer costs in practice.

> **Data boundary.** The sample CSV and 75-phrase workbook are useful for checking that the software runs. We need a separate, independently labelled test set before reporting research accuracy or claiming that any model is better.

Our main question is straightforward: can we improve the Hinglish cases missed by the original BERT/RoBERTa models without losing the useful word-and-emoji focus of the base paper? The final answer will come from the controlled comparisons above, especially negative-class recall and sarcasm errors.